

## Quantum Mechanics:

- Particle of mass  $m$  confined to a ring of radius  $r$  in the  $x-y$  plane

Note  $\phi \sim \phi + 2\pi$

$$x = r \cos(\phi), \quad y = r \sin(\phi)$$

$$\frac{1}{2} m \dot{v}^2 = \frac{1}{2} m \left( \left( \frac{dx}{dt} \right)^2 + \left( \frac{dy}{dt} \right)^2 \right)$$

$$= \frac{1}{2} m r^2 \dot{\phi}^2, \quad V=0$$

$$\therefore \mathcal{L} = T - V = \frac{1}{2} m r^2 \dot{\phi}^2 = \frac{1}{2} I \dot{\phi}^2, \quad I = m r^2$$

$$\therefore S_{\text{free}} = \int dt \frac{1}{2} I \dot{\phi}^2$$

- Now couple this to have charge  $e$ , then we have

$$\mathcal{L} = \int dt \frac{1}{2} I \dot{\phi}^2 + e \oint A_\mu dx^\mu =$$

$$\rightarrow e \oint A_\mu \frac{dx^\mu}{dt} dt, \quad A \equiv \frac{\vec{B}}{2\pi} d\phi, \quad A = A_\phi, \quad A_2 = A_{\text{rco}}$$

$$\mathcal{L} = \int dt \left( \frac{1}{2} I \dot{\phi}^2 + e \frac{\vec{B}}{2\pi} \phi \right)$$

- The second term turns out to be a topological term. Also known as the " $\theta$ "-term. This is due to the fact that it is a total derivative

## Classical symm:

- System will therefore have an  $O(2)$  symmetry.

take  $e^{i\phi}$  on  $S^1$  then the rotation by  $R(\alpha): e^{i\phi} \mapsto e^{i\alpha} e^{i\phi} \rightsquigarrow \phi \mapsto (\phi + \alpha) \bmod(2\pi)$

$$R(\alpha) = \begin{pmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{pmatrix} \quad O(2) \cong U(1)$$

$R$  acts on  $x, y$  coords.  $U(1)$  on  $(r, \phi)$

- The map  $h: \mathbb{R}^+ \mapsto U(1)$ ,  $h(t) := e^{i\phi(t)}$

so we are dealing with 0+2 dim field theory

Note  $S_t = \int dt \int d^d x \mathcal{L}(\phi, \partial \phi)$

~ Non-linear sigma model

Note, For  $U(1)$ : Lie alg.  $U(1) = i\mathbb{R}$

$$A \mapsto g^{-1}(d + A)g = \bar{e}^{i\theta} i d\theta e^{i\theta} + A = A + i d\theta$$

• Symmetry of E.O.M.: Parity transfo.

$$\phi \rightarrow -\phi, \quad \beta A \text{ not invariant (doesn't matter)}$$

• Algebra:  $R_\alpha R_\beta = R_{\alpha+\beta}, P^2 = 1, P R_\alpha P = R_{-\alpha}$

$$\mathcal{O}(2) \cong SO(2) \ltimes \mathbb{Z}_2$$

Field theory: Spacetime Mtd  $M$ ,  $\phi = \phi^a$  on  $M$   
valued in some target space  $X$ .

$$\rightarrow M = \mathbb{R}, \quad X = S^2, \quad u: M \rightarrow S^2$$

Note, the coupling can be written as

$$\oint \in \phi^*(A). \text{ For } \phi: \mathbb{R} \rightarrow S^2 \quad (M \rightarrow X)$$

$$\mathcal{L} = \frac{1}{2} \dot{\phi}^2 + e \frac{\beta}{2\pi} \dot{\phi}, \quad \pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi} + \frac{e\beta}{2\pi}$$

$$\mathcal{H} = \pi \dot{\phi} - \mathcal{L} = (\dot{\phi} + \frac{e\beta}{2\pi}) \dot{\phi} - \frac{1}{2} \dot{\phi}^2 - \frac{e\beta}{2\pi} \dot{\phi}$$

$$\mathcal{H} = \frac{1}{2} \dot{\phi}^2 \quad \therefore \text{only } T \text{ left. without losing } \pi$$

we can write this as

$$H = \int dt \frac{1}{2\pi} (\pi - \frac{e\beta}{2\pi})^2$$

~ Canonical Quant.

$$\{\phi, \pi\} = i$$

$$\pi \rightarrow -i \frac{\partial}{\partial \phi}$$

$$H = \int dt \left( \frac{1}{2\pi} \right) \left( -i \frac{\partial}{\partial \phi} - \beta \right)^2$$

$$\beta := \frac{e\beta}{2\pi\hbar}$$

~ Note

$$\int dt \frac{e\beta}{2\pi} \dot{\phi}$$

$\theta$ -term

\* obtain a family of Ham/Hs  
parametrized by  $\beta$ .

$\theta$ -terms in (2+1)-dim field theory:

Here we take target to be  $\psi \in L^2(\mathbb{R})$



$$i\hbar \frac{\partial}{\partial t} \psi = \frac{\hbar^2}{2I} \left( -i \frac{\partial}{\partial \phi} - \beta \right) \left( -i \frac{\partial}{\partial \phi} - \beta \right) \psi$$

$$= \frac{\hbar^2}{2I} \left( -\frac{\partial^2}{\partial \phi^2} + 2i\beta \frac{\partial}{\partial \phi} + \beta^2 \right) \psi$$

$$i \frac{\partial}{\partial t} \psi = \frac{\hbar^2}{2I} (p^2 - 2\beta p) \psi$$

$$\Leftrightarrow \ln(\psi) = \int dt \frac{\hbar}{2I} (p^2 - 2\beta p)$$

$$\therefore \psi = \exp\left(\int dt \frac{\hbar}{2I} p^2 - 2\beta p\right)$$

$$\psi(\phi) = \int \frac{dp}{\sqrt{2\pi\hbar}} e^{+ip\phi} e^{\int dt \left( \frac{\hbar}{2I} p^2 - 2\beta p \right)}$$

$$= \int \frac{dp}{\sqrt{2\pi\hbar}} e^{ip\phi} \left( 1 + \int dt \frac{\hbar}{2I} p^2 - 2\beta p \right)$$

$$= \delta(\phi - 0) + \int dt dp e^{ip\phi} (p^2 - 2\beta p)$$

$$\hat{H}|\psi\rangle = \lambda|\psi\rangle$$

$$\psi_m = c e^{im\phi}$$

$$\frac{\hbar^2}{2I} \left( i \frac{\partial}{\partial \phi} + \beta \right) \left( i \frac{\partial}{\partial \phi} + \beta \right) \psi$$

$$\frac{\hbar^2}{2I} \left( -\frac{\partial^2}{\partial \phi^2} + 2i\beta \frac{\partial}{\partial \phi} + \beta^2 \right) \psi = \lambda \psi$$

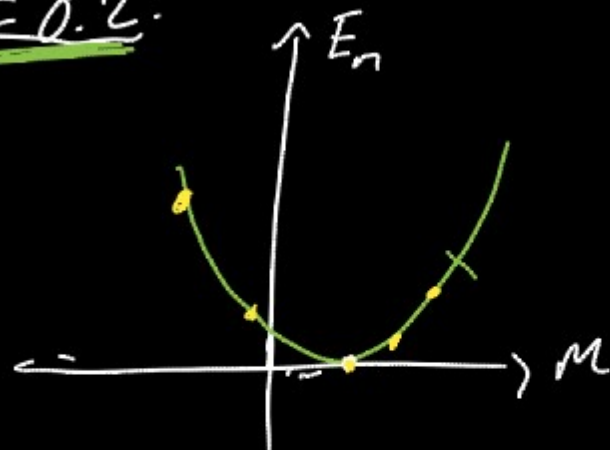
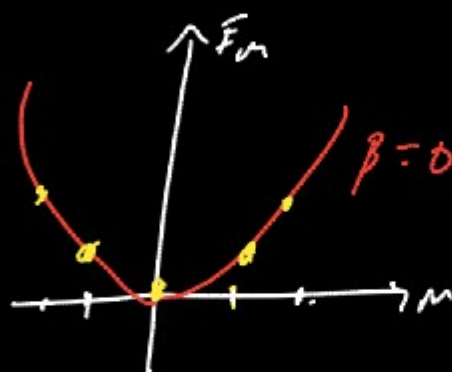
$$\rightarrow \psi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi} \quad m \in \mathbb{Z}$$

$$\text{Energy eigenstates } E_m = \frac{\hbar^2}{2I} (m - \beta)^2$$

plot  $(E_m, m)$

$$(i) \quad \underline{\beta = 0}$$

$$(ii) \quad \underline{\beta = 0.2}$$



$$(iii) \quad \beta = 1/2$$

$$\beta = 1/2$$



$\psi(x)$  has a symmetry under  $m \mapsto 2\beta - m$  ✓  
 $E_m = \frac{\hbar^2}{2I} (m - \frac{1}{2})^2 = \frac{\hbar^2}{2I} (2\beta - m - \beta)^2 = \frac{\hbar^2}{2I} (m - \frac{1}{2})^2$   
 $\therefore$  There is only a symmetry if  
 $2\beta \in \mathbb{Z} \Rightarrow m \mapsto 2\beta - m$  symmetry

(a)  $\theta$ -term has no toroidal effect in the quantum physics:  $\beta$  shifts the spectrum of the quantum ham.

(b) If  $2\beta = 2n$ ,  $n \in \mathbb{Z}$  then the energy eigenspaces are two fold degenerate, except for  $m=0$ :

proof:  $E_m = \frac{\hbar^2}{2I} (m - \tilde{\beta})^2$ ,  $\tilde{\beta} = n \in \mathbb{Z}$   
 $\therefore E_m = \frac{\hbar^2}{2I} (m - n)^2$

$m \mapsto 2\beta - m$ :

$E_m = \frac{\hbar^2}{2I} (2\beta - m - \beta)^2$

$E_0 = 0$ ,  $(m - 2n)^2 = 0 \Rightarrow m = 2n$

• If odd  $2\beta = 2n + 1$

$E_0 = (m - 2n + 1)^2 = 0$

$m = 2n - 1$  or  $m = 2n + 1$

•  $\frac{d}{dn} E_m = \frac{\hbar^2}{2I} \frac{d}{dn} (m - \beta)^2 = 2(m - \beta) \frac{\hbar^2}{2I} = 0$

$m = \beta$

$\frac{d}{d\beta} E_m = 0 \Rightarrow \beta = m$

$m_0$

$$U\psi_m := \psi_{m+1}$$

$$U(\psi_m) = e^{i\varphi} e^{im\varphi} \frac{1}{\sqrt{2\pi}}$$

$$\Leftrightarrow e^{i\varphi} H_B e^{-i\varphi} = H_{B+1}$$

(d) Viewing the system as a field theory:

$$M^2 \rightarrow X:$$

$$S = \int d^{d+1}x \sqrt{|g|} h^{ab}(\sigma) \frac{1}{2} m g_{\mu\nu}(\phi(x)) \partial_a \phi^\mu \partial_b \phi^\nu$$

$$\bullet d\phi: T_x M \rightarrow T_{\phi(x)} X$$

$$\text{we can define } \|d\phi\|^2: \int_M \text{Tr}(d\phi^\dagger d\phi) \text{Vol}(h)$$

$$\Rightarrow d\phi \text{ is a section of } (TM)^* \otimes T^*(TX)$$

$$\text{then } \int_M \|d\phi\|^2$$

$\rightarrow$   $\theta$ -term: has a 1+1-dim analogue, called the B-field term. also obtained from KK-reduction of 1+1-dim Maxwell:

$$S = \frac{1}{2} \int dx^0 dx^1 F_{01}^2 + \frac{\theta}{2\pi} \int F_{01} dx^0 \wedge dx^1$$

$$= \frac{1}{2} \int F \wedge *F + \frac{\theta}{2\pi} \int F$$

$\bullet A_0 = 0$ , only gauge invariant quantity is

$$\exp(i\oint A) = \exp(i\oint_{S^1} A_1 dx^1)$$

$$\therefore \theta = 2\pi \beta$$

$$\bullet \text{Internal symm: } O(2) \quad SO(2) \cong U(1)$$

$\rightarrow$  Quantum:  $SO(2)$  shift symmetry  $\phi \rightarrow \phi + \alpha$

is realised by a translation operator

$$\rho(R(\alpha)): \text{rep of } R(\alpha) \text{ on } \mathcal{H}_m$$

$$\rho(R(\alpha)) = R(\alpha): \mathcal{H}_m \xrightarrow{\sim} \mathcal{H}_m$$

$$[R(\alpha)\psi_m] := e^{im\alpha}\psi_m$$

• We can also find a rep  $\rho$  for the parity op.

$$\rho(P), \text{ Name: } P H_0 P^{-1} = H_{-B}$$

Note, for general values of  $\beta$  the operator  $H_\beta$  is not unitarily eq. to  $H_0$  i.e.  $\exists P$  which push us to  $H_{-\beta}$ .

\* Only when  $2\beta \in \mathbb{Z}$  and so  $\phi \mapsto -\phi$  takes  $m \mapsto -m$  should be accompanied by  $(e^{i\kappa\phi})^{2\beta} = e^{i2\beta\phi}$

so,

$P$ : maps  $\psi_m \mapsto \psi_{2\beta-m}$

$$(P\psi_m) := \sum_n \psi_{2\beta-m} \quad \text{is: } \text{phase: } \sum_n \pm 1$$

$$P = (U^{2\beta} \circ e^{-i\kappa\phi}) = \begin{pmatrix} e^{i2\beta\phi} & e^{-i2\beta\phi} \end{pmatrix}$$

$$\begin{aligned} PR(\alpha)P\psi_m &= PR(\alpha)\psi_{2\beta-m} = P e^{i(2\beta-m)(\phi+\alpha)} \\ &= e^{i(2\beta-m)\alpha} \psi_m \\ &= e^{2i\beta\alpha} e^{-i\alpha m} \psi_m = e^{2i\beta\alpha} R(-\alpha) \end{aligned}$$

$\therefore$

$$PR(\alpha)P = e^{2i\beta\alpha} R(-\alpha)$$

$$\{P, R(\alpha), z \in \mathbb{Z}\} : \alpha \sim \alpha + 2\pi, z \in \mathbb{C}$$

$$\text{Let } G_\beta := \{P, R(\alpha), z \in \mathbb{Z}\}$$

\* Naturally we would expect  $G_\beta \cong O(2) \times U(1)$

however we note that

$$PR(\alpha)P = e^{2i\beta\alpha} R(-\alpha)$$

is not satisfied by a simple direct product

It will be an extension of

$$O(2) \times U(1) \rightarrow \mathbb{C}^* \rightarrow 1$$

(i) If  $2\beta \in \mathbb{Z}$ , or, even stronger, if  $\beta \in \mathbb{Z}$ , then  
 $\exists$  an isomorphism between  $C_p$  and  $U(1) \times O(2)$

$$\tilde{R}(\alpha) := e^{-i\beta\alpha} R(\alpha)$$

(ii) If  $\beta \in \frac{1}{2}\mathbb{Z}$ , then  $\tilde{R}$  is not well defined  
 since  $e^{-\frac{1}{2}n\alpha} \rightarrow e^{-\frac{1}{2}n\alpha} e^{-\frac{\alpha}{2}}$  or  $e^{-\frac{n\alpha}{2}} e^{-\frac{\alpha}{2}}$   
 for different  $n$  and  $\beta = \frac{1}{2}n$ .

$\rightarrow$  To understand what happens to the theory when  
 a  $\theta$ -field term is included we work w/  $\text{Spin}(2)$ .

The double covering group of  $O(2)$ .

$$\text{Spin}(2) \cong \text{SU}(2)/\mathbb{Z}_2 \cong \text{U}(1) \cong \mathbb{R}/\mathbb{Z}$$

Consider a  $\text{Spin}(2)$  parametrization

$$\tilde{\alpha} \mapsto \tilde{\alpha} + 2\pi$$

as a matrix rep of  $\text{Spin}(2)$ , define  $\hat{R}(\tilde{\alpha})$ .

The matrix group is generated by the  
 spinor algebra elements. Here

$$\hat{R}(\tilde{\alpha}) = \exp(\tilde{\alpha} \sigma^1 \sigma^2) = \cos(\tilde{\alpha}) + i \sin(\tilde{\alpha}) \sigma^3$$

which has a double cover

$$\hookrightarrow \pi: \text{Spin}(2) \rightarrow \text{SO}(2)$$

$$\hookrightarrow \pi: \text{SU}(2) \rightarrow \text{SO}(3)$$

$$\text{and } \mathfrak{spin}(2) \subset \text{SU}(2)$$

\* Note  $\hat{R}(\tilde{\alpha})$  is generated by the chirality  
 (matrix  $\gamma$  clifford element  $\gamma_* = (-i)^{D+1} \gamma_0 \dots \gamma_{D-1}$   
 here  $\gamma_* = \gamma_0 \gamma_1 \gamma_2 = \sigma^1 \sigma^2$

group about  $\mathbb{Z}$ .

$$\tilde{\pi}: \tilde{R}(\alpha) \rightarrow R(2\alpha)$$

$\mathbb{Z}_2$  then acts on  $\text{Spin}(2)$  via an outer automorphism.  $\mathbb{Z}_2$  has the operator/matrix rep. of a parity operator.

Homomorphism

$$\hat{\pi}: \mathbb{Z}_2 \rightarrow \text{Aut}(\text{Spin}(2))$$

Defined by

$$\hat{\pi}(\hat{p}): \text{Spin}(2) \rightarrow \text{Spin}(2)$$

$$(\hat{\pi}(\hat{p}))(\tilde{R}(\alpha)) := \tilde{R}(\alpha)^{-1} = \tilde{R}(-\alpha)$$

$\Rightarrow$  Define the Pin group as the semidirect product

$$\text{Pin}^+(2) \cong \text{Spin}(2) \rtimes_{\hat{\pi}} \mathbb{Z}_2$$

$\text{Spin}(n)$ : Double cover of  $\text{SO}(n)$

$\text{Pin}(n)$ : " "  $\text{O}(n)$

If  $2\beta = (2n+1)$ ,  $\mathfrak{g}_{\beta}$  is generated by

$$\pm 1_{\mathbb{Z}}$$

$$\rho(\tilde{R}(\alpha)) = e^{-2\beta\alpha} \tilde{R}(\alpha)$$

$$\rho(\hat{p}) = p$$

$$1 \rightarrow \mathbb{Z}_2 \rightarrow \text{Pin}^+(2) \rightarrow \text{O}(2) \rightarrow 1$$

$$\downarrow \quad \quad \downarrow \rho \quad \quad \downarrow 11$$

$$1 \rightarrow \text{U}(1) \rightarrow \mathfrak{g}_{\beta} \rightarrow \text{O}(2) \rightarrow 1$$

$$\mathbb{Z}_2 \cong \{\pm 1\} \subset \text{U}(1)$$

We note:

(i) The classical theory has an  $O(2)$  symm.

(ic) The Quantum theory has broken  $SO(2)$  symm  
if  $2\beta \notin \mathbb{Z}$ .

(iic) Quantum:  $2\beta \in 2\mathbb{Z}$ , then  $\exists O(2)$  symm

$$1 \rightarrow U(1) \rightarrow G_\beta \rightarrow O(2) \rightarrow 1$$

$$G_\beta / U(1) \cong O(2) \quad \text{or} \quad G_\beta \cong U(1) \times O(2)$$

(iiv) Quantum:  $2\beta = 2\mathbb{Z} + 1$

$$1 \rightarrow U(1) \rightarrow G_\beta \rightarrow O(2) \rightarrow 1$$

Does NOT split as  $U(1) \times O(2)$

and must implement the  $\tilde{Pin}(2)$  group: A central extension of  $O(2)$  via  $\mathbb{Z}_2$ .

\* Important Note: The charged scalar field has spin 0. However the representation space of this particle was "lifted" to a spinor rep of the classical symmetry group. So the nontrivial topology / inclusion of  $B$ -terms ( $\theta$ -terms) has caused a 'curious' shift. (That's cool.)

Quantum statistical mechanics of the charged particle on a ring:

$$Z = \text{Tr}_\mathcal{H} (e^{-\beta H}) \quad , \quad \beta = \frac{1}{kT}$$

Since we have diagonalized  $H$  into energy eigenvalues:

$$Z = \sum_{m \in \mathbb{Z}} e^{-\frac{\beta}{2I} (m - \beta)^2}$$

(i) The expression is periodic under integer shifts of  $\beta$ . Theory is invariant under integral shifts of the "theta-angle"  $\beta$ .

(ii) At low temp,  $\beta \rightarrow \infty$ , we only have a single dominant term in the sum which is  $m = B$

or if  $2\beta = 2\mathbb{Z} + \mathbb{I}$ , then the groundstate is a Qbit.

~> Take the IR limit to understand the ground state.

Restate  $Z = \int_0^{2\pi} d\phi \langle \phi | e^{-\beta H} | \phi \rangle$

Since the trace  $\text{Tr} \rightarrow \int_0^{2\pi} d\phi \langle \phi |$

\* Interpret  $\langle \phi | e^{-\beta H} | \phi \rangle$  as a specialization of the matrix elements of Euclidean time prop.

$$\langle \phi_2 | e^{-t H_E} | \phi_1 \rangle = \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} e^{-\frac{\beta E}{2\pi} (n - B)^2 + i n (\phi_2 - \phi_1)}$$

$\phi_2 = \phi_1 = \phi$ : taking Euclidean time to be periodic w/ period  $\beta$ :  $\phi(0) = \phi(\beta)$

$$Z(\phi_1, \phi_1 | \beta) = \langle \phi_1 | e^{-\beta H} | \phi_1 \rangle = \int [D\phi(t)]_{\phi(0)=\phi_1}^{\phi(\beta)=\phi_2} e^{-\frac{1}{2\pi} \int_0^\beta dt}$$

$$\exp \left( -\frac{1}{2\pi} \left( \int_0^\beta dt + \frac{1}{2} \int_0^\beta dt \dot{\phi}^2 - i \int_0^\beta dt B \dot{\phi} \right) \right)$$

Note: E.O.M:  $\ddot{\phi} = 0$

$$\phi(t) = c + b t, \quad \phi(0) = \phi_1 = a$$

$$\phi(\beta) = \phi_2 = \phi_1 + b\beta$$

$$\phi_a(t) = \phi_1 + \left( \frac{\phi_2 - \phi_1 + 2\pi\omega}{\beta} \right) t, \quad \omega \in \mathbb{Z}$$

$$e^{\phi_a(t)} = \exp \left[ i \left( \left( 1 - \frac{t}{\beta} \right) \phi_1 + \frac{t}{\beta} \phi_2 \right) + \frac{2\pi i t \omega}{\beta} \right]$$

## Instanton solutions

- \* There are two circles which are are dealing w/  
The spacetime of (0+1)-dim is the Euclidean circle.  
the target space is also a circle.

$$S^1_E \xrightarrow{\phi} S^1$$

$$e^{i\phi} : M \rightarrow X, \quad e^{i\phi}(t) := e^{i\phi(t)}$$

$$\pi^1(S^1) \cong \mathbb{Z}$$

$\Leftrightarrow \text{Hom}(M, X)$  has different connected components.

- The different top. sectors are uniquely labeled by the winding # of  $e^{i\phi(t)}$ .

In the path int we sum over all field conf.

- We can now expand  $\phi$  around  $\phi_{cl}$

$$\phi = \phi_{cl} + \phi_q$$

where  $\phi_q$  are the quantum fluctuations, with

$$\phi_q(0) = \phi_q(\beta) = 0$$

- $\Rightarrow \phi_q$  is topologically a trivial component of  $\text{Hom}(M, X)$   
since  $\phi_q$  is a loop at 0.

$$\begin{aligned} \rightarrow & \frac{1}{2} \int_0^\beta \frac{1}{2} I (\dot{\phi}_{cl} + \dot{\phi}_q)^2 - i \int \beta d\phi (\dot{\phi}_{cl} + \dot{\phi}_q) \\ & = S_{cl} + S_q \end{aligned}$$

$$Z(\phi_1, \phi_2 | \beta) = Z_q \sum_{w \in \mathbb{Z}} e^{-\frac{2\pi i F}{\rho} \left( w + \frac{\phi_2 - \phi_1}{2\pi} \right)^2 + 2\pi i \beta \left( w + \frac{\phi_2 - \phi_1}{2\pi} \right)}$$

$$Z_q = \int [d\phi_q(t)]_{\phi_q(0)=0}^{\phi_q(\beta)=0} e^{-\int_0^\beta \frac{1}{2} I \dot{\phi}_q^2}$$

Proof:

$$\begin{aligned} Z(\phi_1, \phi_2 | \beta) &:= \int [d\phi(t)]_{\phi(t)=\phi_1}^{\phi(t)=\phi_2} e^{-S_{cl}} e^{-S_q} \\ &= \sum_{w \in \mathbb{Z}} e^{-S_{cl}} \int [d\phi_q(t)]_{\phi_q(0)=0}^{\phi_q(\beta)=0} e^{-S_q} \end{aligned}$$

$$= \frac{1}{\beta} \exp \left( -\frac{1}{\beta} \int_0^\beta dt \left( \frac{1}{2} \dot{\phi}^2 + \frac{1}{2} \phi^2 \right) \right) = \frac{1}{\beta} \exp \left( -\frac{1}{\beta} \int_0^\beta dt \left( \frac{1}{2} \dot{\phi}^2 + \frac{1}{2} \phi^2 \right) \right)$$

$$\begin{aligned} &= \frac{1}{\beta} \left( \phi_2 - \phi_1 + 2\pi\omega \right)^2 \frac{1}{2\beta} + i\beta \left( \phi_2 - \phi_1 + 2\pi\omega \right) \\ &= \left( -\frac{1}{\beta} \right) 2\pi^2 \left( \omega + \frac{\phi_2 - \phi_1}{2\pi} \right)^2 + 2\pi i \beta \left( \phi_2 - \phi_1 + \omega \right) \\ &\therefore \sum_{\omega \in \mathbb{Z}} \exp \left( \frac{2\pi^2}{\beta} \left( \omega + \frac{\phi_2 - \phi_1}{2\pi} \right)^2 + 2\pi i \beta \left( \phi_2 - \phi_1 + \omega \right) \right) \end{aligned}$$

$$\int [d\phi_g(t)]_{\phi_g(0)=0}^{\phi_g(\beta)=0} e^{-\int_0^\beta dt \frac{1}{2} \dot{\phi}^2} \equiv \mathbb{Z}_g$$

$$\rightarrow \text{Note } \int_0^\beta dt \dot{\phi}_g = \phi_g \Big|_{t=0}^{t=\beta} = 0$$

\* Note: We are integrating over "all" maps  $\phi_g$   
 $\phi_g: [0, \beta] \rightarrow U(1)$ ,  $\phi_g(0) = \phi_g(\beta) = 0$   
 so they are homotopically trivial.

$$\int \left( \prod_{i=1}^n \frac{dx^i}{\sqrt{2\pi}} \right) e^{-\frac{1}{2} x^i A_{ij} x^j + b_i x^i} = \frac{1}{\sqrt{\det A}} e^{\frac{1}{2} b_i (A^{-1})^{ij} b_j}$$

$$\int [d\phi_g] e^{-\int_0^\beta dt \phi_g \left( -\frac{1}{2} \frac{d^2}{dt^2} \phi_g \right)}$$

$\mathcal{O} \equiv \left( -\frac{1}{2} \frac{d^2}{dt^2} \right)$  is the operator of interest.

$$= (2\pi) (\text{Det}(\mathcal{O}))^{-1/2}$$

\*  $\zeta$ -f<sup>n</sup> regularization: only works for trace-class ops.

$$\frac{d}{ds} \Big|_{s=0} \lambda^{-s} = \log(\lambda)$$

$$\text{proof } \frac{d}{ds} \lambda^{-s} = \lambda^{-s} - \log(\lambda) \Big|_{s=0} = -\log(\lambda)$$

Define

$$\zeta_{\mathcal{A}}(s) = \sum_{\lambda} \lambda^{-s}$$

This sum is over the spectrum of  $\mathcal{A}$ .

If  $\mathcal{A}$  is diagonalizable:

$$\prod_{\lambda} \lambda = \exp(-\zeta'_{\mathcal{A}}(0))$$

Proof:  $\frac{d}{ds} \zeta_{\mathcal{A}}(s) = \sum_{\lambda} \lambda^{-s} (-\log(\lambda))$

$$-\zeta'_{\mathcal{A}}(0) = \sum_{\lambda} \log(\lambda)$$

$$\exp\left(\sum_{\lambda} \log(\lambda)\right) = \prod_{\lambda} e^{\log(\lambda)} = \prod_{\lambda} \lambda$$

• Define  $\boxed{\text{Det}(\mathcal{A}) \equiv \exp(-\zeta'_{\mathcal{A}}(0))}$

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Gauging the Global  $SO(2)$  symmetry and Chern-Simons  
Terms:

Steps:

(i) Make the sym local and equivariant by coupling

(ii) Integrate over the Gauge fields.

\* If you view field theory as a functor  
 $\mathcal{F}: \mathcal{B}_{(n-1,n)}^{str} \rightarrow \mathcal{T} \leftarrow \text{tensor cat.}$

$\rightarrow$  Borsom of  $n-1, n$  - dim mds.

(i)  $\mathcal{F}^{eff}: \mathcal{B}^{str} + \text{spin. trace w/can} \rightarrow \mathcal{T}$

cto Change Bundles to Bundles w/ connection.

(ii) Sum over isomorphism classes

Take  $\phi(t) \mapsto \phi(t) + \alpha(t)$ . Here

$\int \dot{\phi}^2$  is not invariant under this trans. we introduce the U(1) Gauge field

$A^{(e)}(t) \neq A_m$  from the solenoid.

$A^{(e)}(t) \mapsto A^{(e)}(t) + i g^{-1} dt g$

$g = C^{i\alpha(t)}$

$A^{(e)}(t) \mapsto A^{(e)}(t) - \partial_t \alpha(t)$

$\therefore A^{(e)} \in C^\infty(M) \otimes \Omega^2(M)$

$A^{(e)} = A^{(e)}(t) dt$

$A^{(e)}$  connection 1-form on  $SO(2)$  Principle Bundle.

We can turn this into a "standard" Gauge theory

Recall, that  $\phi$  is defined on  $S^1$

so we can define a complex scalar field,

invariant (Globally) by

$\Phi(t) = e^{i\phi(t)}$

$S = \int \frac{1}{2} \mathcal{F}(\dot{\phi} + A^{(e)})^2 dt + \oint \mathcal{B}(\dot{\phi} + A^{(e)}) dt$

$$+ \oint B E(t)^{-1} \left( -\frac{i}{c} + A^{(e)} \right) E(t) dt$$

This is invariant under

$$\phi(t) \rightarrow \phi(t) + \alpha(t)$$

$$A^{(e)} \rightarrow A^{(e)}(t) - \partial_t \alpha(t)$$

[The following text is a dense, continuous block of characters and symbols, appearing to be a corrupted or heavily redacted document. It contains no legible words or phrases.]